Section C: Interpretation & Visualisation

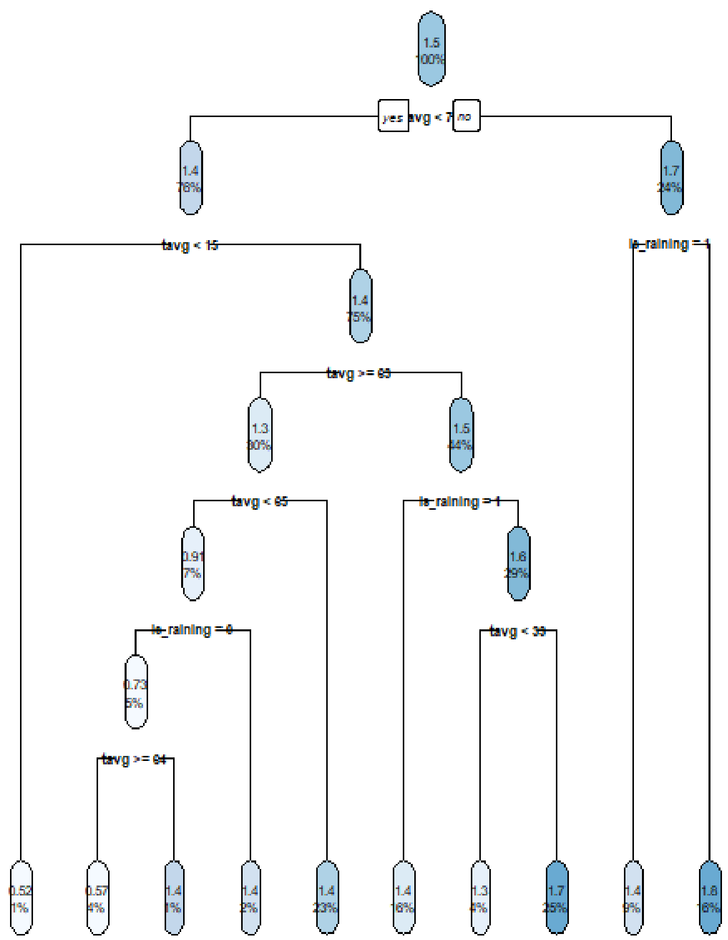
Part 1: Product summaries

Note: Whilst is_weekend was initially included in my code as a control variable, it does not show a direct relationship between climate patterns and therefore has been removed from the weather analysis below.

Product 5

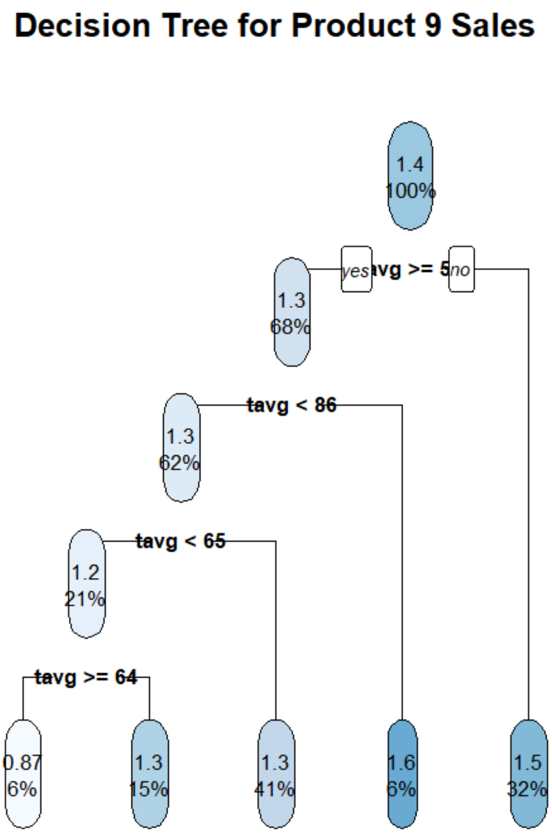
- According to my linear regression model, temperature has a statistically significant, positive relationship with units_sold with an estimated coefficient of 0.0038. This explains a small but steady increase in volume as the temperature gets warmer.
- Precipitation heavily decreases demand with a negative coefficient of -0.1995 for is_raining. Furthermore there is a primary split at 77° F where optimal log sales(1.7) take place during warm, rainy conditions.
- However, the model's low validation R2 score confirms the idea that weather is a secondary factor influencing units_sold.

Decision Tree for Product 5 Sales



Product 9

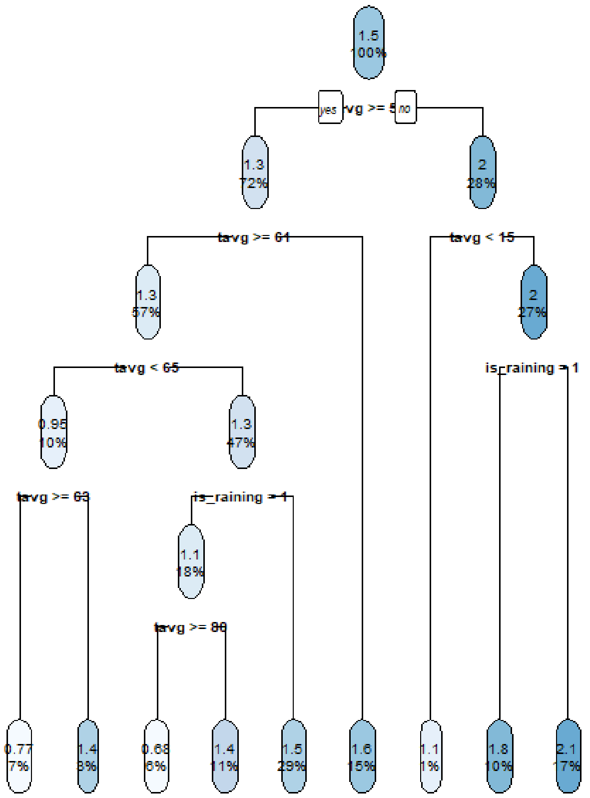
- Compared to the other products, product 9 has a unique relationship with weather variables
- It experiences a slight negative relationship with a coefficient of -0.0033 for average but reacts positively to precipitation with a coefficient of +0.0770.
- This indicates that rainy days spark a minor increase in purchases for product 9. A potential reason for this increase may be that product 9 is more inconvenient in wet weather.
- The decision tree suggests that the sweet spot for product 9 purchases lie between 54° F and 86° F as the average log sales are at their highest of 1.5 and 1.6.
- Product 9 represents a slight contradiction however, the linear model says sales go down slightly as the temperature becomes warmer but the decision tree displays that the highest sales happen at warm temperatures between 54° F to 86° F. The linear model misses this sweet spot in the warm temperature as it can only draw a straight line here but the decision tree splits the data into blocks and captures this analysis nicely.



Product 45

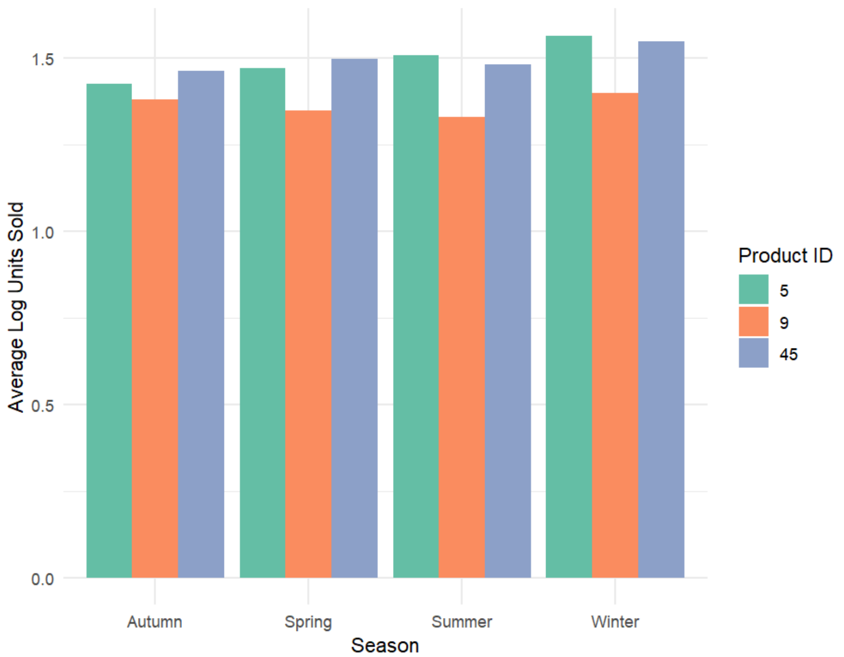
- This product demonstrates the most powerful relationship with environmental variables
- The linear regression model identifies the temperature as a major negative driver with a coefficient of -0.0132.
- Additionally, rain severely deters customer spending yielding a heavy coefficient of -0.2518 for is_raining.
- The regression tree tells us that when conditions drop below 51° the predicted log sales node jumps from 1.3 to 2.0 showing a major demand for product 45 in cold seasonal weather.

Decision Tree for Product 45 Sales



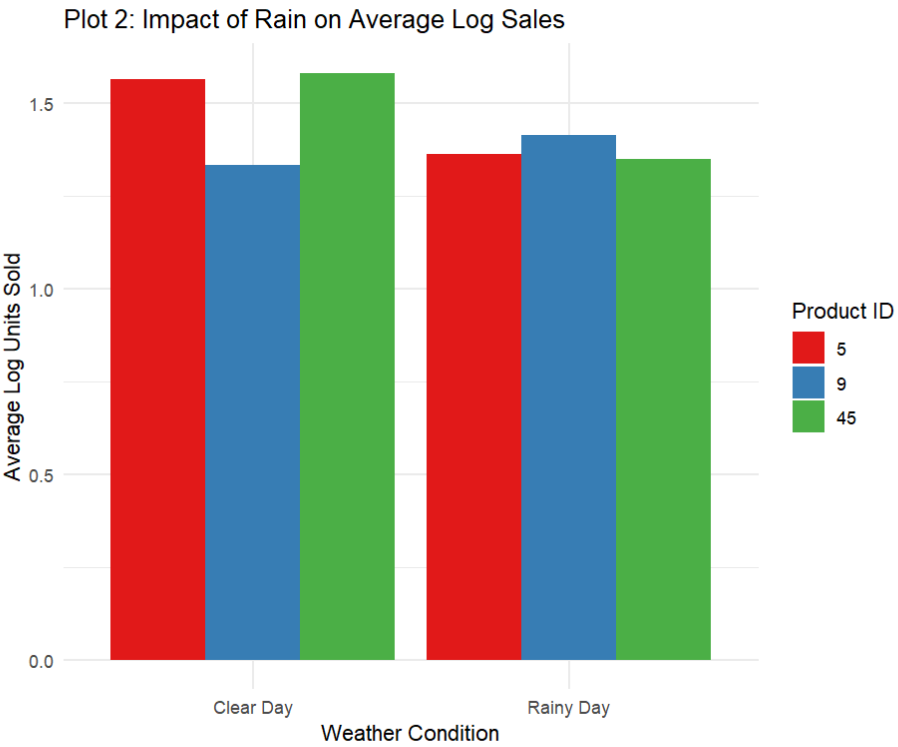
Part 2 Visualisations

Plot 1: Average Log Sales Across Seasons

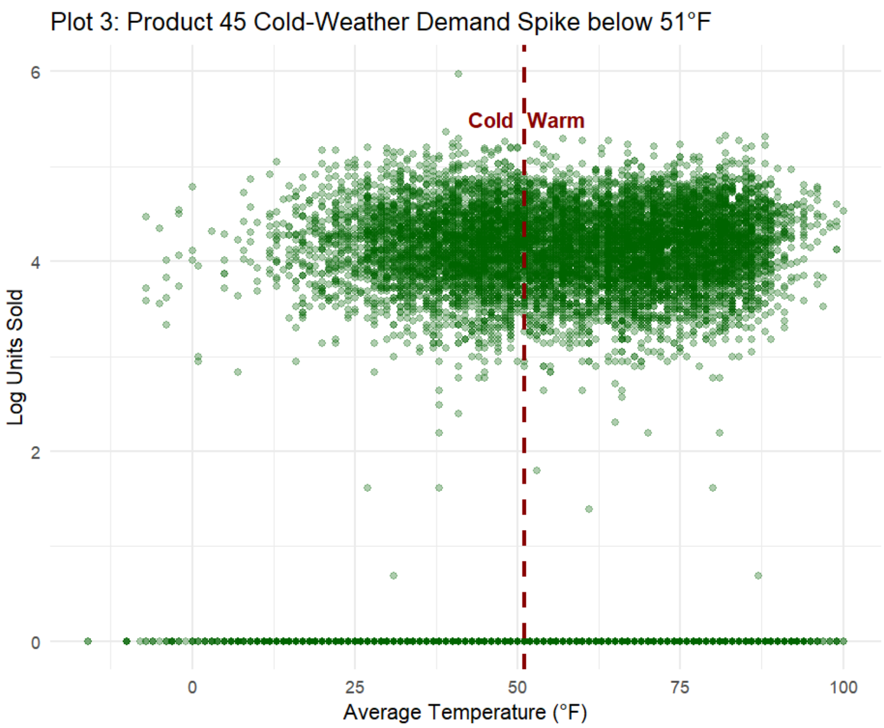


- This clustered bar graph displays the average log units for the top 3 products grouped by the 4 seasons.

- This plot validates the seasonal coefficients of our model, it shows product 45s dominance during Autumn and Winter(matching the splits for cold weather in the decision tree) and product 5s slight elevation in the warmer months.



- This bar chart contrasts the average log sales of each product on clear days verses rainy days
- This graph provides visual proof of the sales drop for product 5 and 45 during rainy days whilst illustrating the slight upward sales rise for product 9 on rainy days.



- The continuous scatter plot displays daily log sales for product 45 across every recorded temperature and is split between warm and cold temperature.
- This plot provided evidence for our top regression tree node split. The split data cloud shows that below 51 ° F, sales compress upward between log 4.0 and 5.0 whereas the Warm side exhibits a downward sag and higher variance in units sold.

Part 3 Business Recommendations

Approximately double the stock for Product 45 below temperatures of 51° F

- Supply chain managers should implement an automatic inventory trigger at 51° F
- This is justified with the regression tree model which has a large difference in log sales when the temperature is above or below 51 ° F, the difference between 1.3 and 2.0 in log values represents an approximate double in raw units stock.
- They should boost local store stock by approximately 100% as soon as weather forecasts predict temperature to be below 51° F.

Decrease outdoor promotions for products 5 and 45 on Rainy days by approximately 20%

- This is due to the precipitation feature. Is_raining produced a coefficient of -0.1995. My data is log transformation so -0.20(rounded) translated to an approximate 20% decrease in units_sold.
- The money spent on promoting product 5 and 45 outdoors in these instances should be redirected to digital campaigns or shifting stock indoors to match the lower shopping volumes.

Increase stock of product 9 in rainy weather by approximately 7.7%

- This is justified with the is_raining coefficient of +0.0770 meaning that sales volume increases on rainy days by approximately 7.7%.

Increase stock of product 9 when temperatures transition from cold to warm by approximately 10%

- This is because the regression tree model shows an increase in log sales value from 1.5 to 1.6 when the weather changes from cold (less than 54° F) to warm.